



Understanding the active-site nature of vanadia-based catalysts for oxidative dehydrogenation of ethylbenzene with CO₂ via atomic layer deposited VO_x on γ-Al₂O₃

Guo-Qing Yang^{a,1}, Huan Wang^{a,1}, Ting Gong^b, Yong-Hong Song^a, Hao Feng^{b,*}, Han-Qing Ge^a, Hui-bin Ge^a, Zhao-Tie Liu^a, Zhong-Wen Liu^{a,*}

^a Key Laboratory of Syngas Conversion of Shaanxi Province, School of Chemistry & Chemical Engineering, Shaanxi Normal University, Xi'an 710119, China

^b State Key Laboratory of Fluorine and Nitrogen Chemicals, Xi'an Modern Chemistry Research Institute, Xi'an 710065, China

ARTICLE INFO

Article history:

Received 16 July 2019

Revised 26 September 2019

Accepted 6 October 2019

Available online 31 October 2019

Keywords:

Vanadium oxide

Atomic layer deposition

Active site

Oxidative dehydrogenation

Ethylbenzene

Carbon dioxide

ABSTRACT

Supported vanadium oxides are widely used as active catalysts for a number of oxidation and oxidative dehydrogenation reactions, and the understanding the structure-function relationship is of great importance for developing more efficient vanadium oxide catalysts. In this work, atomic layer deposition (ALD) was utilized to disperse sub-monolayer VO_x over a commercial γ-Al₂O₃ support. Characterization results indicate that highly dispersed VO_x species with different structures were obtained by simply changing the number of ALD cycles. The vanadium oxides synthesized by performing 1 cycle of ALD were exclusively isolated monomeric VO_x. With increasing the number of ALD cycles from 3 to 8, the polymeric VO_x species were formed, and the polymerization extent or domain size of VO_x was increased. On the contrary, crystalline V₂O₅ besides polymeric VO_x was found if the number of ALD cycles was extended to 12. Catalytic performances of ALD VO_x/γ-Al₂O₃ were evaluated for the oxidative dehydrogenation of ethylbenzene with CO₂ (CO₂-ODEB), and the isolated monomeric VO_x species were found to be more active and more stable than the polymeric VO_x species and crystalline V₂O₅. By correlating the specific activity and the structural characteristics of ALD VO_x/γ-Al₂O₃ catalysts, the V–O–Al bonds were concluded as the key active sites for CO₂-ODEB.

© 2019 Elsevier Inc. All rights reserved.

1. Introduction

The most important application of vanadium besides metallurgical use is in catalysis, and supported vanadium oxides (VO_x) catalyze many industrially important reactions [1,2]. The original purpose of loading VO_x on oxide supports is to increase the surface area and mechanical strength of the catalyst [1]. However, later studies reveal that the activity of the supported VO_x is strongly influenced by the specific type of oxide support [1–3], which is attributed to the metal oxide-support-interaction effect. Although many efforts have been made by using theoretical calculations and various characterization techniques, the key issue related to the reactivity of the catalyst still remains unresolved [4–6]. As a

result, the rational design of vanadium based catalysts is still a great challenge. Using Raman spectroscopy, three types of surface VO_x species, i.e., isolated VO_x, polymerized VO_x, and crystalline V₂O₅, have been clearly distinguished in supported VO_x catalysts [7,8]. However, the formation and distribution of different VO_x species are reciprocally affected by the synthesis method of the catalyst, the loading or surface density of VO_x species, and the nature of the support [9–11]. Thus, as the lacking of a supported vanadium catalyst with a well-distributed specific structure of VO_x species, it is still difficult to identify unambiguously its active nature for oxidation and oxidative dehydrogenation reactions.

Styrene (ST) is one of the most important monomers for polymer industries [12]. In comparison with the industrial route for the production of ST via the catalytically direct dehydrogenation of ethylbenzene (EB), the oxidative dehydrogenation of EB to ST with CO₂ (CO₂-ODEB) is an efficient, energy-saving, and environmentally friendly process [13–15]. Supported VO_x catalysts with excellent activity are widely studied for CO₂-ODEB [15,16]. However, the common problem for CO₂-ODEB is the severe deactivation of catalysts, which has been the bottleneck for the industrialization

* Corresponding authors at: School of Chemistry & Chemical Engineering, Shaanxi Normal University, No. 620, West Chang'an Avenue, Xi'an 710119, China (Zhong-Wen Liu); Xi'an Modern Chemistry Research Institute, Xi'an 710065, China (H. Feng).

E-mail addresses: fenghao98@hotmail.com (H. Feng), zwliu@snnu.edu.cn (Z.-W. Liu).

¹ These authors contributed equally to this work.